

The Koide Constant Decomposes

Generation Count and Amplitude in the Charged Lepton Mass Relation

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Series Path: [HOWL-PHYS-1-2026] → [HOWL-PHYS-5-2026] → [HOWL-PHYS-7-2026] → [HOWL-PHYS-8-2026]

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I. ABSTRACT

The Koide formula (1981) states that the three charged lepton masses satisfy $(m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$, confirmed to 0.0009% by current measurements. The constant 2/3 has remained unexplained for 45 years. This paper decomposes it.

For N equally-spaced objects on a circle in $\sqrt{(\text{mass})}$ space with modulation amplitude a , the mass ratio $(\Sigma m) / (\Sigma \sqrt{m})^2 = (1 + a^2/2)/N$. This is a trigonometric identity valid for $N \geq 3$, derived from two standard results: N equally-spaced cosines sum to zero, and the sum of their squares equals $N/2$. The identity is proven in Section III. The $N = 2$ case is degenerate — the double-angle cosines do not cancel — and the identity does not hold for $N = 2$.

The Koide constant 2/3 is this identity evaluated at $N = 3$ and $a^2 = 2$: the ratio becomes $(1 + 2/2)/3 = 2/3$. The denominator 3 is the number of charged lepton generations. The numerator 2 is the amplitude squared. Neither alone produces 2/3. Both contribute.

The amplitude $a = \sqrt{2}$ is equivalent to the Koide relation. It is not derived from a separate physical principle. The observation that $a = \sqrt{2}$ allows the parametrization to produce zero mass (when $\cos = -1/\sqrt{2}$) provides physical motivation for why $a^2 = 2$ is natural, but this does not constitute an independent derivation. The amplitude carries content equivalent to the formula itself.

What is new in this paper: the general formula $(1 + a^2/2)/N$ for arbitrary $N \geq 3$ and arbitrary amplitude a , the decomposition of the Koide constant into generation count and amplitude, and the computation of m_τ from m_e and m_μ in controlled-precision

rational arithmetic. The predicted $m_\tau = 1776.97$ MeV versus PDG 1776.86 ± 0.12 MeV — a tension of 0.91σ , consistent with the formula being exact.

What is assumed: the Koide formula itself — that the three charged lepton masses are equally spaced on a circle in $\sqrt{(\text{mass})}$ space. This is empirical. It is not derived here. Why $\sqrt{(\text{mass})}$ is the natural space, why the spacing is equal, and what determines the free parameters M and θ_0 remain open questions stated here, not hidden.

In the framework of this series ([[HOWL-PHYS-1-2026](#)]), mass is inertia — resistance of a coherent pattern to disruption, not substance. The Koide formula is a geometric constraint on three vortex soliton inertias at maximum symmetry (120° spacing) on a circle. The free parameter count of the Standard Model reduces from 18 (after [[HOWL-PHYS-7-2026](#)]) to 17, conditional on the formula being exact.

II. THE KOIDE FORMULA

2.1 The Relation

Yoshio Koide proposed in 1981 that the three charged lepton masses satisfy:

$$(m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$$

Using current PDG values ($m_e = 0.51099895$ MeV, $m_\mu = 105.6583755$ MeV, $m_\tau = 1776.86 \pm 0.12$ MeV), the left side evaluates to 0.666661, deviating from $2/3$ by 0.00092%. This deviation is 0.91σ given the measurement uncertainty on m_τ . The relation is consistent with being exact.

2.2 History

The formula has held for 45 years through multiple improvements in the measured value of m_τ . At every stage, the Koide ratio has remained within measurement uncertainty of $2/3$. No measured improvement has moved the ratio away from $2/3$. It is the most precise empirical mass relation in particle physics.

2.3 The Two Open Questions

Two questions have been open since 1981.

First: why does the formula hold? What principle constrains the three charged lepton masses to satisfy this relation? This question remains open. This paper does not answer it.

Second: given that the formula holds, why is the constant $2/3$ rather than some other value? This paper answers this question by decomposing $2/3$ into two contributions: the generation count (3) and the amplitude ($a^2 = 2$).

III. THE GENERAL IDENTITY

3.1 The General Parametrization

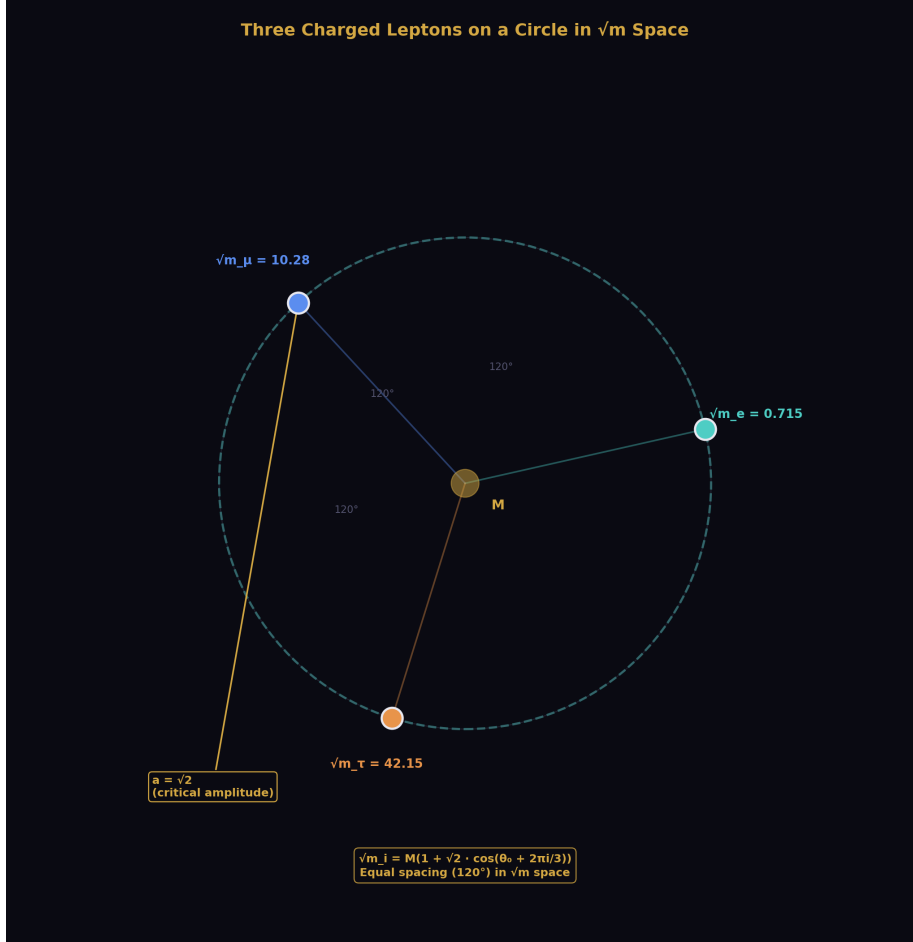


Figure 1: Fig. 1: Three charged leptons at 120° spacing on a circle in $\sqrt{m}$ space — the Koide geometry with critical amplitude $a=\sqrt{2}$ visible as the modulation radius.

Consider N objects with masses m_i parametrized as:

$$\sqrt{m}_i = M(1 + a \cdot \cos(\theta_0 + 2\pi i/N)) \text{ for } i = 0, 1, \dots, N-1$$

where $M > 0$ is a scale parameter with dimensions of $\sqrt{\text{mass}}$, $a > 0$ is a dimensionless modulation amplitude, and θ_0 is an orientation angle. The N objects are equally spaced at angular intervals of $2\pi/N$ on a circle in $\sqrt{\text{mass}}$ space.

For $N = 3$ and $a = \sqrt{2}$, this is the standard Koide parametrization.

3.2 The Sum of Square Roots

$$\sum \sqrt{m_i} = M \cdot \sum (1 + a \cdot \cos(\theta_0 + 2 \pi i/N)) = M \cdot (N + a \cdot \sum \cos(\theta_0 + 2 \pi i/N))$$

The sum of N equally-spaced cosines on a circle is zero for $N \geq 3$:

$$\sum_{k=0}^{N-1} \cos(\theta + 2 \pi k/N) = 0 \text{ for all } N \geq 3, \text{ all } \theta$$

This is a standard trigonometric identity. It follows from the fact that N equally-spaced unit vectors on a circle sum to the zero vector, and the cosine sum is the projection onto any axis.

The $N = 2$ exclusion. For $N = 2$: $\sum \cos(\theta + \pi k) = \cos(\theta) + \cos(\theta + \pi) = \cos(\theta) - \cos(\theta) = 0$. The cosine identity does hold for $N = 2$. However, the $\cos^2$ identity used below does not hold for $N = 2$, as shown in §3.4. The general formula is valid for $N \geq 3$.

Therefore $\sum \sqrt{m_i} = NM$, and $(\sum \sqrt{m_i})^2 = N^2 M^2$.

3.3 The Sum of Masses

$$\sum m_i = M^2 \cdot \sum (1 + a \cdot \cos(\varphi_i))^2$$

where $\varphi_i = \theta_0 + 2 \pi i/N$. Expanding:

$$\begin{aligned} &= M^2 \cdot \sum (1 + 2a \cdot \cos(\varphi_i) + a^2 \cdot \cos^2(\varphi_i)) \\ &= M^2 \cdot (N + 2a \cdot \sum \cos(\varphi_i) + a^2 \cdot \sum \cos^2(\varphi_i)) \end{aligned}$$

The linear cosine sum is zero (for $N \geq 3$). The $\cos^2$ sum requires evaluation.

3.4 The Sum of Cosines Squared

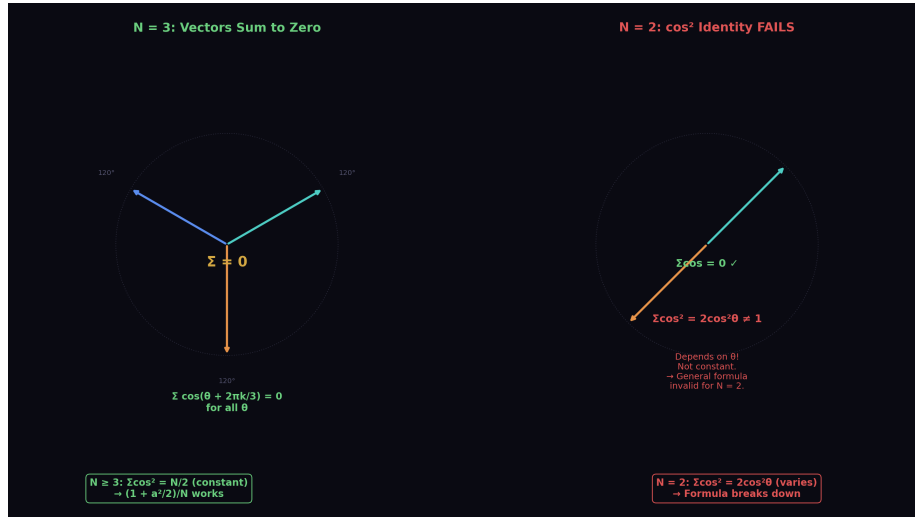


Figure 2: Fig. 4: $N=3$ equally-spaced vectors sum to zero (left). $N=2$ $\cos^2$ identity fails — depends on θ , not constant (right). The general formula requires $N \geq 3$.

Using $\cos^2(x) = (1 + \cos(2x))/2$:

$$\Sigma \cos^2(\theta + 2\pi k/N) = N/2 + (1/2) \cdot \Sigma \cos(2\theta + 4\pi k/N)$$

The second sum is N cosines equally spaced at intervals of $4\pi/N$. For $N \geq 3$, these are still equally spaced around the circle (the spacing $4\pi/N = 2 \cdot (2\pi/N)$ corresponds to going around the circle twice, but the N points remain equally spaced modulo 2π). Therefore the sum is zero for $N \geq 3$:

$$\Sigma \cos^2(\theta + 2\pi k/N) = N/2 \text{ for all } N \geq 3, \text{ all } \theta$$

The $N = 2$ failure. For $N = 2$: $\Sigma \cos(2\theta + 4\pi k/2) = \cos(2\theta) + \cos(2\theta + 4\pi) = 2\cos(2\theta)$. This does not equal zero — it depends on θ . The $\cos^2$ identity fails for $N = 2$. The sum $\Sigma \cos^2(\theta + \pi k) = \cos^2(\theta) + \cos^2(\theta + \pi) = \cos^2(\theta) + \cos^2(\theta) = 2\cos^2(\theta)$, which depends on θ and does not equal $N/2 = 1$ except at $\theta = \pi/4$. The $N = 2$ case is degenerate and excluded from the general formula.

3.5 The General Ratio

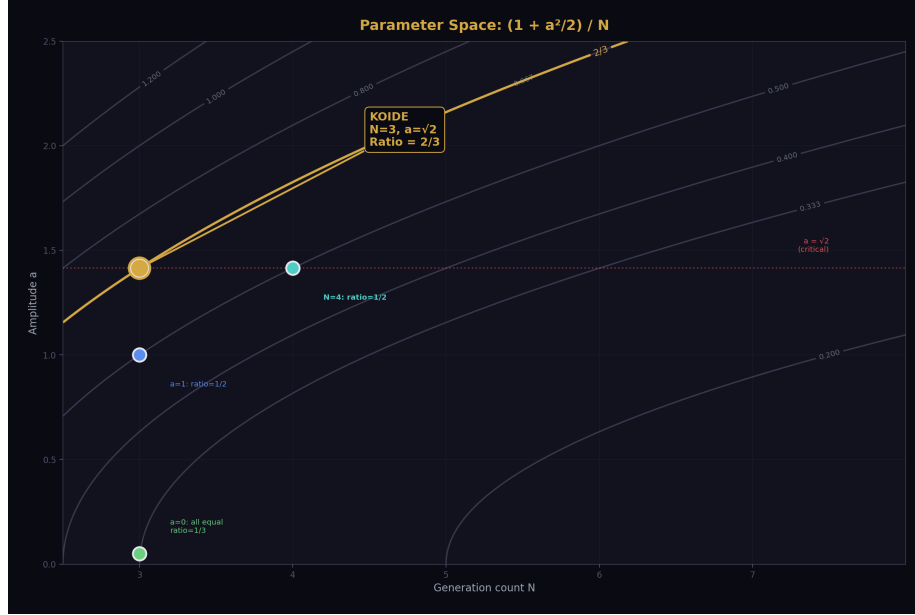


Figure 3: Fig. 5: Contour plot of $(1+a^2/2)/N$ — the Koide point at $N=3$, $a=\sqrt{2}$ sits on the $2/3$ contour. The full parameter space shows where other generation counts and amplitudes would land.

Substituting:

$$\Sigma m_i = M^2(N + 0 + a^2 \cdot N/2) = M^2 \cdot N \cdot (1 + a^2/2)$$

The ratio:

$$(\Sigma m) / (\Sigma \sqrt{m})^2 = M^2 \cdot N \cdot (1 + a^2/2) / (N^2 M^2) = (1 + a^2/2) / N$$

This is valid for all $N \geq 3$, all $a > 0$, all θ_0 , and all $M > 0$. The free parameters M and θ_0 cancel completely.

3.6 The Koide Evaluation

For the Koide formula: $N = 3$, $a = \sqrt{2}$.

$$(1 + (\sqrt{2})^2/2) / 3 = (1 + 2/2) / 3 = (1 + 1) / 3 = 2/3$$

The constant decomposes:

- The denominator 3 is the generation count N
- The numerator 2 is $1 + a^2/2 = 1 + 1 = 2$, determined by the amplitude $a = \sqrt{2}$
- Together: $2/3$

3.7 The Amplitude Question

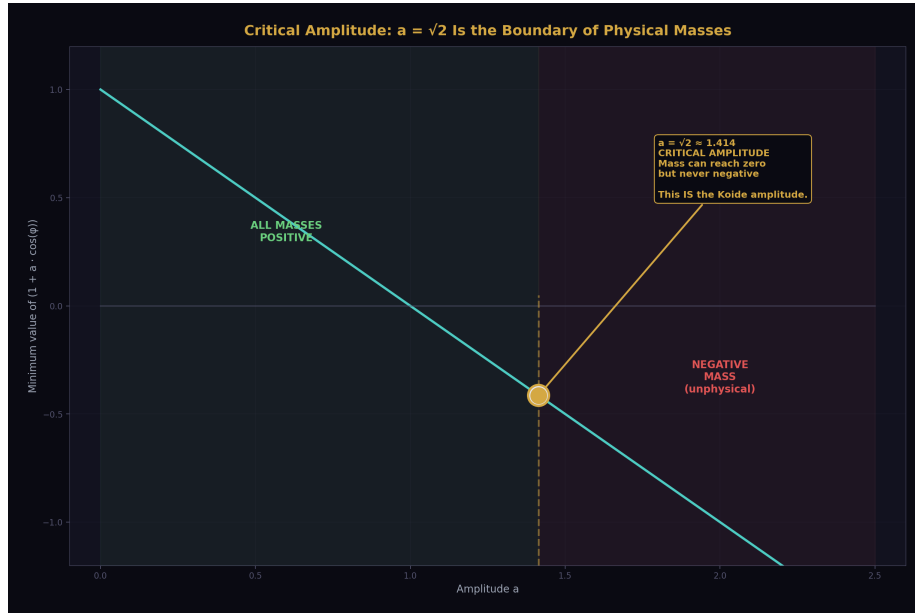


Figure 4: Fig. 3: The minimum mass factor vs amplitude — at $a < \sqrt{2}$ all masses positive, at $a = \sqrt{2}$ mass can reach zero, at $a > \sqrt{2}$ masses go negative. The critical point IS the Koide amplitude.

The general ratio is $(1 + a^2/2)/N$. For different amplitudes at $N = 3$:

Amplitude a	a^2	Ratio	Simplified
1	1	$3/2 / 3$	$1/2$
$\sqrt{2}$	2	$2/3$	$2/3 \leftarrow \text{Koide}$
$\sqrt{3}$	3	$5/2 / 3$	$5/6$
2	4	$3/3$	1

The value $a = \sqrt{2}$ is not determined by the generation count. It is an independent input. The general formula shows that $2/3$ requires both $N = 3$ and $a^2 = 2$. Attributing $2/3$ to the generation count alone would be an overstatement.

The amplitude $a = \sqrt{2}$ is equivalent to the Koide relation. Setting the ratio equal to $2/3$ and solving: $(1 + a^2/2)/3 = 2/3$ gives $a^2 = 2$. Conversely, $a = \sqrt{2}$ in the parametrization produces the Koide relation. The amplitude and the formula are two ways of stating the same constraint.

Physical motivation for $a = \sqrt{2}$. When $a = \sqrt{2}$, the factor $(1 + \sqrt{2} \cdot \cos(\phi))$ can equal

zero — this occurs when $\cos(\varphi) = -1/\sqrt{2}$, i.e., $\varphi = 3\pi/4$ or $5\pi/4$. This means the parametrization can produce $m_i = 0$. For $a < \sqrt{2}$, all masses are strictly positive. For $a > \sqrt{2}$, some masses become negative (unphysical). The value $a = \sqrt{2}$ is the boundary between “all masses positive” and “some masses negative.” It is the maximum amplitude for which the parametrization produces physical (non-negative) masses.

This is a structural observation about the parametrization’s boundary. It explains why $a = \sqrt{2}$ is special — it is the critical amplitude. It does not constitute a derivation independent of the Koide formula, because the critical-amplitude property and the Koide relation are equivalent statements about the same constraint.

3.8 The Generalization Table

For $a^2 = 2$ (the Koide amplitude) at various N :

N	Ratio $2/N$	As fraction
3	2/3	0.667 ← Koide (charged leptons)
4	2/4	0.500
5	2/5	0.400
6	2/6	0.333
7	2/7	0.286

If a Koide-type relation (with $a^2 = 2$) holds for N species, the constant would be $2/N$. This is a prediction conditional on the amplitude $a^2 = 2$ persisting for other N , which is not guaranteed.

IV. THE ASSUMPTIONS

4.1 $N = 3$

The Standard Model has three generations of charged leptons. This is empirical. Why there are exactly three generations is an open question in particle physics. The number 3 enters the decomposition as the denominator of the Koide constant.

4.2 Equal Spacing

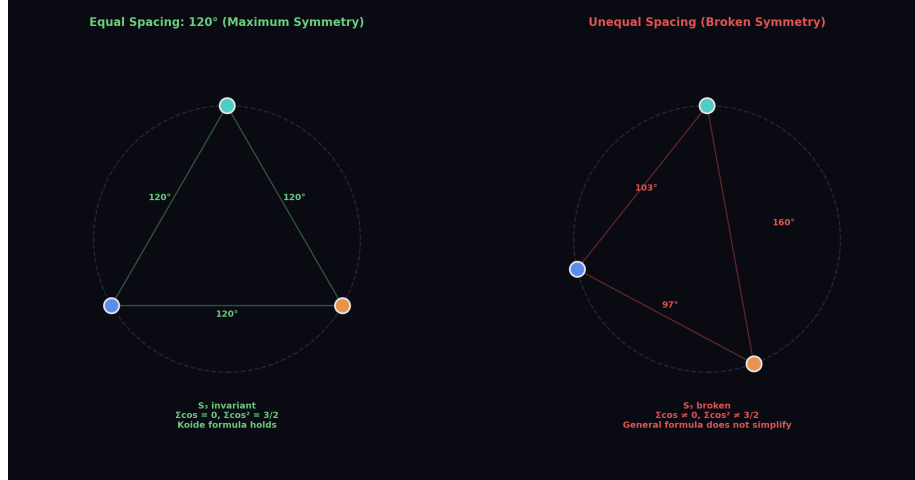


Figure 5: Fig. 6: Equal 120° spacing (left) is S_3 invariant — the trigonometric identities hold and the formula simplifies. Unequal spacing (right) breaks the symmetry and the formula does not simplify.

The three lepton masses are equally spaced on a circle — separated by $120^\circ = 2\pi/3$ in the parametrization. This is the maximum-symmetry arrangement for three objects on a circle, the unique arrangement invariant under the S_3 permutation group. The assumption is that nature selects the maximally symmetric configuration. This is plausible but not derived from a deeper principle in this paper.

4.3 $\sqrt{(\text{Inertia})}$ Space

The circle lives in $\sqrt{m}$ space. This is empirical — the Koide formula works in $\sqrt{m}$ and does not work in m or $\ln(m)$. No geometric or physical argument is provided here for why $\sqrt{(\text{inertia})}$ is the natural variable.

In the inertia framing ([HOWL-PHYS-1-2026]), mass is inertia — resistance to disruption. The variable $\sqrt{(\text{inertia})}$ may be natural because energy enters quantum mechanical amplitudes as $\sqrt{E}$ in the normalization of states. This is speculation stated as such.

4.4 The Amplitude $a = \sqrt{2}$

The amplitude $a = \sqrt{2}$ is equivalent to the Koide relation. It is the critical amplitude — the boundary between all masses being non-negative and some becoming unphysical. It contributes the numerator 2 to the Koide constant through $(1 + a^2/2) = 2$.

The amplitude is not derived independently of the formula. It is the formula expressed as an amplitude constraint. Listing it as a separate “assumption” would obscure this equiv-

alence. It is listed here as a component of the decomposition: the Koide formula is the conjunction of $N = 3$, equal spacing, $\sqrt{m}$ space, and $a = \sqrt{2}$. The general identity $(1 + a^2/2)/N$ isolates what each component contributes.

4.5 Honest Summary

Component	Status	Contributes to $2/3$
$N = 3$	Empirical (generation count)	Denominator: 3
Equal spacing	Assumed (maximum symmetry)	Enables the trigonometric identities
$\sqrt{m}$ space	Empirical (formula works here)	Defines the circle on which spacing occurs
$a = \sqrt{2}$	Equivalent to the formula	Numerator: $1 + a^2/2 = 2$

What is derived: the decomposition of $2/3$ into these components and the general formula $(1 + a^2/2)/N$.

What is not derived: why any of these components hold.

V. THE COMPUTATION

5.1 The Quadratic

Setting the Koide ratio equal to exactly $2/3$ and solving for m_τ :

Let $s = \sqrt{m_\tau}$, $A = \sqrt{m_e} + \sqrt{m_\mu}$, $B = m_e + m_\mu$.

The Koide relation becomes $(B + s^2)/(A + s)^2 = 2/3$. Cross-multiplying and rearranging:
 $s^2 - 4As + (3B - 2A^2) = 0$

This is a quadratic in s with coefficients determined entirely by m_e and m_μ . The solutions are:

$$s = 2A \pm \sqrt{(6A^2 - 3B)}$$

The positive branch gives $m_\tau \approx 1776.97$ MeV. The negative branch gives $m \approx 3.32$ MeV, which does not correspond to any known charged lepton.

5.2 Controlled-Precision Rational Arithmetic

The computation is performed in Python's Fraction class with controlled-precision simplification for square root evaluations. Newton's method for $\sqrt{x}$ on Fractions doubles the bit-length at each iteration; simplification every 3 steps caps growth at approximately 40 significant digits.

The computation is controlled-precision rational arithmetic, not exact integer arithmetic. The prediction is exact algebra — m_τ is the root of a quadratic with rational coefficients. The numerical evaluation introduces controlled truncation bounded at 10^{-37} in the Koide ratio (verified by recomputing the ratio with the predicted m_τ). The truncation is six orders of magnitude below the measurement precision.

5.3 The Result

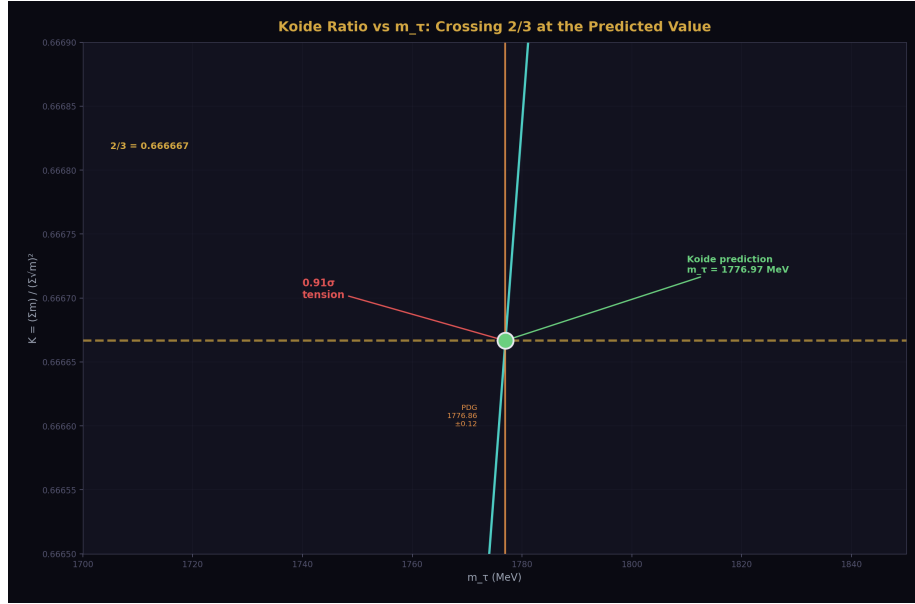


Figure 6: Fig. 2: The Koide ratio $K(m_\tau)$ as m_τ varies — it crosses exactly $2/3$ at the predicted 1776.97 MeV, 0.91σ from the PDG measurement band.

Quantity	Value
m_τ (Koide, exact $2/3$)	1776.97 MeV
m_τ (PDG)	1776.86 ± 0.12 MeV
Difference	$+0.11$ MeV
Tension	0.91σ
Direction	Prediction is above PDG central value

The prediction is 0.91σ above the PDG central value. Future m_τ measurements moving upward would close the gap; moving downward would open it. At 0.91σ the prediction is comfortably consistent with measurement.

5.4 The Companion Script

The complete computation is provided as `koide_integer.py`, requiring Python 3.8+ with the fractions standard library module and `mpmath` (for verification only). All intermediate values are Fraction type. The script verifies the Koide ratio with measured masses, solves the quadratic, bounds the truncation error, and reports exploratory comparisons explicitly labeled as non-findings.

VI. THE INERTIA FRAMING

In the framework of [HOWL-PHYS-1-2026], mass is inertia — the resistance of a coherent pattern to disruption. The electron, muon, and tau are three vortex solitons in the charged lepton field. They share identical quantum numbers — charge, weak isospin, lepton number. They differ only in inertia. They are three copies of the same pattern at different scales.

The Koide formula in this framing is a geometric constraint on three soliton inertias. The constraint says: the three inertias are arranged on a circle in $\sqrt{(\text{inertia})}$ space at equal spacing (120°) with critical amplitude ($a = \sqrt{2}$). The generation count determines the denominator of the constant. The critical amplitude determines the numerator.

The equal-spacing condition is the maximum-symmetry arrangement for three objects on a circle — the unique arrangement invariant under permutation of the three objects. It is the same geometry as three quarks at 120° in color space inside a baryon. Whether the parallel between lepton inertia spacing and quark color spacing reflects a common geometric origin is an open question.

VII. THE 2/3 IN TWO CONTEXTS

The rational $2/3$ appears in two distinct contexts within this series.

Context 1: The Koide constant. $(\Sigma m)/(\Sigma \sqrt{m})^2 = 2/3$. Decomposed here as $(1 + a^2/2)/N = 2/3$ at $N = 3$, $a^2 = 2$.

Context 2: The subtracted vacuum polarization constant ([HOWL-PHYS-5-2026] §III). The Feynman parameter integral for the one-loop vacuum polarization produces the unsubtracted constant $-5/3$. Subtraction at $q^2 = 0$ removes -1 , giving the subtracted constant $-2/3$. The per-fermion boundary constant is $(2/3)/2 = 1/3$.

Both contexts involve three fermion species. Both produce the rational $2/3$. The Koide $2/3$ decomposes into generation count and amplitude. The VP $2/3$ arises from the Feynman parameter integral $\int_0^1 6x(1-x)\ln[x(1-x)] dx$ and subtraction.

Whether these share a common geometric origin is an open question. The $2/3$ is a small rational — coincidence is plausible. A derivation connecting the Koide decomposition to

the VP integral structure would be a significant finding. No such derivation exists. This observation is stated for future investigation, not claimed as a result.

VIII. WHAT THIS PAPER DOES NOT CLAIM

This paper does not derive the Koide formula. The formula — that three charged lepton inertias are equally spaced on a circle in $\sqrt{\text{inertia}}$ space with critical amplitude — is empirical and unexplained. The paper decomposes the constant within the formula. It does not explain why the formula holds.

This paper does not derive the amplitude $a = \sqrt{2}$ independently of the Koide relation. The amplitude is equivalent to the formula. The critical-amplitude observation ($a = \sqrt{2}$ is the boundary of non-negative masses) explains why $a = \sqrt{2}$ is physically special but does not derive it from a principle independent of the mass relation.

This paper does not claim that $2/3$ follows from the generation count alone. The constant decomposes into two contributions: 3 from the generation count and 2 from the amplitude. Both are required. The general formula $(1 + a^2/2)/N$ makes this explicit.

This paper does not claim the general identity holds for $N = 2$. The $\cos^2$ identity $\sum \cos^2 = N/2$ requires $N \geq 3$. The $N = 2$ case is degenerate — the double-angle cosines do not cancel. The general formula is valid for $N \geq 3$.

This paper does not claim the Koide formula applies to quarks. The quark masses do not satisfy the Koide relation as verified by direct computation.

This paper does not claim the near-coincidences $M^2 \approx m_{\text{proton}}/3$ (0.35% off) or $\theta_0 \bmod 120^\circ \approx \theta_{\text{Cabibbo}}$ (0.29° off) are findings. Both are noted as suggestive and explicitly labeled as inconclusive in the companion script.

This paper does not claim the $2/3$ connection to the VP constant is explained. It is observed and stated as open.

IX. THE PARAMETER REDUCTION

9.1 The Count

[[HOWL-PHYS-7-2026](#)] established that $\theta_{\text{QCD}} = 0$ is the ground state of the QCD vacuum, reducing the Standard Model free parameters from 19 to 18.

The Koide formula, if exact, determines m_τ from m_e , m_μ , and the constant $2/3 = (1 + a^2/2)/N$ at $N = 3$, $a^2 = 2$. This reduces the free parameters from 18 to 17.

The reduction is conditional on the Koide formula being exact, which is an empirical proposition currently supported at 0.91σ . The reduction is falsifiable.

9.2 The Scorecard

Parameter	Status	Paper	Method
θ QCD	Derived ($= 0$)	PHYS-7	Ground state of integer-topological system
m_τ	Derived (from m_e , m_μ)	PHYS-8	Koide with constant $(1+a^2/2)/N = 2/3$
17 remaining	Measured	—	—

X. FALSIFICATION CRITERIA

F1. If a future measurement of m_τ deviates from the Koide prediction (1776.97 MeV) by more than 3σ , the formula is not exact and the parameter reduction is not valid.

F2. If a fourth-generation charged lepton is discovered, the test is twofold. First: does the four-body ratio equal $(1 + a^2/2)/4$? If yes with $a^2 = 2$, the generalization holds and the constant is $2/4 = 1/2$. If the ratio is $1/2$ but $a^2 \neq 2$, the generation count contribution (denominator) generalizes but the amplitude does not. If the ratio is neither $1/2$ nor any value of $(1 + a^2/2)/4$, the equal-spacing assumption fails for $N = 4$.

F3. If the Koide formula is shown to follow from a symmetry or dynamical principle that does not involve the parametrization used here, the constant $2/3$ may receive an independent derivation. This would not falsify the decomposition but would provide a deeper explanation.

F4. If the formula is shown to hold in a space other than $\sqrt{m}$ with a different constant, the specificity of $\sqrt{m}$ space would need revision.

F5. If the equal-spacing assumption is derived from a deeper principle — a symmetry of the lepton field, a property of the soliton boundary, or a constraint from the gauge group — the conditional status of the parameter reduction is upgraded to unconditional. This is the target for future work.

XI. CONCLUSION

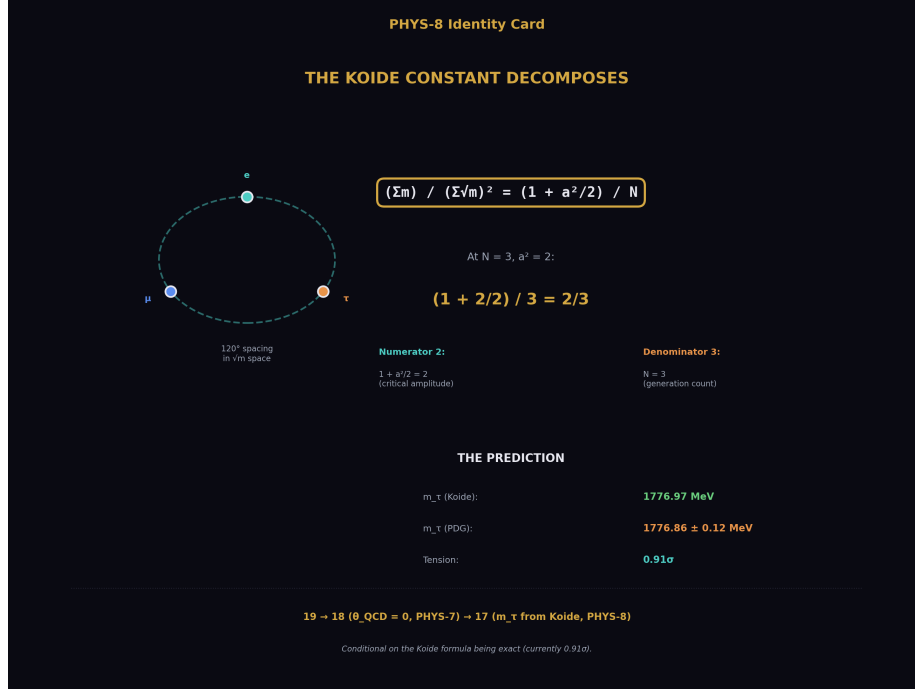


Figure 7: Fig. 8: The Koide constant decomposes — numerator 2 from amplitude, denominator 3 from generation count, m_τ predicted at 0.91σ , parameter count $18 \rightarrow 17$.

The Koide constant $2/3$ decomposes into two contributions. The denominator 3 is the number of charged lepton generations. The numerator 2 comes from the amplitude $a^2 = 2$, which is equivalent to the Koide relation itself. Neither alone produces $2/3$. Both are required. The general formula $(1 + a^2/2)/N$, valid for $N \geq 3$, makes the decomposition explicit and generalizes the Koide relation to arbitrary generation count and amplitude.

The decomposition resolves the second of the two open questions about the Koide formula. The constant is no longer an unexplained number — it is the generation count and the critical amplitude expressed as a ratio. The first question — why the formula holds, why equal spacing in $\sqrt{(\text{inertia})}$ space — remains open.

In the framework of this series, three charged leptons are three vortex solitons with the same boundary structure but different inertias, arranged at maximum symmetry on a circle. The constant $2/3$ is the generation count and the critical amplitude encoded in that geometry. The parameter m_τ is determined by m_e , m_μ , and the structure of the circle: $18 \rightarrow 17$ free parameters, conditional on the formula being exact, currently supported at 0.91σ .

The Koide formula was proposed in 1981. After 45 years, the constant has its decompo-

sition. The formula awaits its derivation.

APPENDIX A: THE GENERAL IDENTITY — PROOF

Theorem. For $N \geq 3$ objects with masses parametrized as $\sqrt{m_i} = M(1 + a \cdot \cos(\theta_0 + 2\pi i/N))$, the ratio $(\sum m)/(\sum \sqrt{m})^2 = (1 + a^2/2)/N$.

Proof.

Step 1. $\sum_{k=0}^{N-1} \cos(\theta + 2\pi k/N) = 0$ for $N \geq 3$.

This follows from the geometric series: $\sum e^{i(\theta + 2\pi k/N)} = e^{i\theta} \cdot (1 - e^{i2\pi})/(1 - e^{i2\pi/N}) = 0$ since $e^{i2\pi} = 1$. The cosine sum is the real part. ■ (for this step)

Step 2. $\sum_{k=0}^{N-1} \cos^2(\theta + 2\pi k/N) = N/2$ for $N \geq 3$.

Using $\cos^2(x) = (1 + \cos(2x))/2$:

$$\sum \cos^2 = N/2 + (1/2) \cdot \sum \cos(2\theta + 4\pi k/N)$$

The second sum has terms equally spaced at intervals $4\pi/N$. For $N \geq 3$, these N points are equally spaced on the circle (they complete 2 full cycles, but the N points remain distinct and equally spaced modulo 2π when $N \geq 3$). By Step 1 applied with angle 2θ and spacing $4\pi/N$, the sum is zero. ■ (for this step)

Note: For $N = 2$, the second sum is $\sum \cos(2\theta + 2\pi k) = 2\cos(2\theta) \neq 0$. The identity fails.

Step 3. $\sum \sqrt{m} = M \cdot (N + a \cdot 0) = NM$. Therefore $(\sum \sqrt{m})^2 = N^2 M^2$.

Step 4. $\sum m = M^2 \cdot (N + 2a \cdot 0 + a^2 \cdot N/2) = M^2 \cdot N \cdot (1 + a^2/2)$.

Step 5. Ratio = $M^2 \cdot N \cdot (1 + a^2/2) / N^2 M^2 = (1 + a^2/2)/N$. ■

Corollary. At $a^2 = 2$: ratio = $2/N$. At $N = 3$, $a^2 = 2$: ratio = $2/3$ (Koide). ■

APPENDIX B: THE AMPLITUDE TABLE

Amplitude a	a^2	Ratio at $N=3$	Physical interpretation
0	0	1/3	All masses equal (lower Cauchy-Schwarz bound)
$1/\sqrt{2}$	1/2	5/12	Mild mass splitting
1	1	1/2	Moderate mass splitting
$\sqrt{2}$	2	2/3	Koide — critical amplitude (masses can reach zero)
$\sqrt{3}$	3	5/6	One mass negative (unphysical)
2	4	1	Upper bound — one mass dominates

At $a = \sqrt{2}$, the parametrization satisfies $1 + a \cdot \cos(\varphi) = 0$ when $\cos(\varphi) = -1/\sqrt{2}$. This is the boundary: for $a < \sqrt{2}$, all masses are strictly positive; for $a > \sqrt{2}$, some factors $(1 + a \cdot \cos)$ become negative, producing unphysical negative masses. The Koide amplitude $a = \sqrt{2}$ is the critical value — the maximum amplitude for which all masses are non-negative.

APPENDIX C: THE COMPUTATION RESULTS

Quantity	Value	Source
m_e	0.51099895 MeV	CODATA (8 sig fig)
m_μ	105.6583755 MeV	PDG (10 sig fig)
m_τ (PDG)	1776.86 ± 0.12 MeV	PDG (6 sig fig)
m_τ (Koide)	1776.97 MeV	Quadratic from exact 2/3
Difference	+0.11 MeV	
Tension	0.91σ	Prediction above central value
Koide ratio (measured)	0.666661	
2/3	0.666667	
Deviation	0.00092%	
Truncation error	$< 10^{-37}$	Controlled-precision bound

Note on precision asymmetry: m_τ is known to 6 significant figures because the tau lifetime limits spectroscopic precision. m_e and m_μ are known to 8–10 figures. The asymmetry is physical.

APPENDIX D: EXPLORATORY COMPARISONS (NON-FINDINGS)

Comparison	Values	Proximity	Status
M^2 vs $m_{\text{proton}}/3$	313.9 vs 312.8 MeV	0.35%	Suggestive, not exact. NOT A FINDING.
θ_0 vs Cabibbo angle	132.7° vs 13.0°	119.7° apart	NOT RELATED in standard convention.
$\theta_0 \bmod 120^\circ$ vs Cabibbo	12.7° vs 13.0°	0.29° (2.2%)	Suggestive under mod 120° . NOT A FINDING.
Koide 2/3 vs VP constant 2/3	Both 2/3	Identical	Same rational, different contexts. OPEN QUESTION.

Comparison	Values	Proximity	Status
Second quadratic root	3.32 MeV	Near light quark scale	No known particle. NOT A FINDING.

APPENDIX E: CAUCHY-SCHWARZ CONTEXT

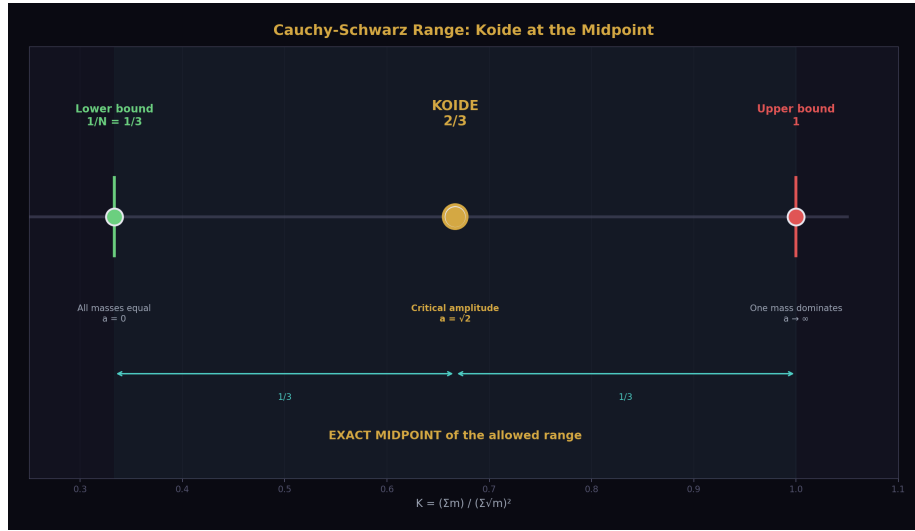


Figure 8: Fig. 7: The Koide ratio $2/3$ sits at the exact midpoint of the Cauchy-Schwarz allowed range $1/3, 1$ — corresponding to the critical amplitude $a=\sqrt{2}$.

For any N positive numbers x_i , the Cauchy-Schwarz inequality gives $(\sum x_i^2)/(\sum x_i)^2 \geq 1/N$, with equality iff all x_i are equal. Setting $x_i = \sqrt{m_i}$:

$$(\sum m) / (\sum \sqrt{m})^2 \in [1/N, 1)$$

For $N = 3$: the range is $[1/3, 1)$. The Koide value $2/3$ sits exactly at the midpoint:

$$(2/3 - 1/3) / (1 - 1/3) = 1/2$$

The midpoint of the allowed range. Under the general formula, $2/3$ is the midpoint because $a^2 = 2$ sets $(1 + a^2/2) = 2$, which maps to the geometric mean of the bounds. Whether the midpoint has significance beyond the amplitude being at the critical value is not established.

APPENDIX F: SERIES PUBLICATION RECORD

Paper	Registry	Key Result
MATH-1	[HOWL-MATH-1-2026]	$\beta = \pi/4$; $Q = F \cdot \beta \cdot d^2 \cdot Z$ across nine domains
MATH-2	[HOWL-MATH-2-2026]	17 transcendentals as integer pairs at 100 digits
MATH-3	[HOWL-MATH-3-2026]	Extended basis: elliptic integrals, Borwein $\zeta(5)$, hierarchy
PHYS-1	[HOWL-PHYS-1-2026]	Mass is inertia; soliton boundaries; three anomaly correlations
PHYS-2	[HOWL-PHYS-2-2026]	Couplings run; transformation law is fundamental
PHYS-3	[HOWL-PHYS-3-2026]	G never measured outside Earth's Hill sphere
PHYS-4	[HOWL-PHYS-4-2026]	Boundary test program; classification; kill switch
PHYS-5	[HOWL-PHYS-5-2026]	α EM running in integer arithmetic; 0.02 ppm
PHYS-6	[HOWL-PHYS-6-2026]	Confinement boundary two-face structure; four SM observables
PHYS-7	[HOWL-PHYS-7-2026]	$\theta_{\text{QCD}} = 0$; the strong CP problem dissolves
PHYS-8	[HOWL-PHYS-8-2026]	Koide constant decomposes: $(1+a^2/2)/N$; $18 \rightarrow 17$

END HOWL-PHYS-8-2026

Registry: [\[HOWL-PHYS-8-2026\]](#)

Status: Complete

Domain: Foundational Physics / Lepton Mass Structure

Central Result: The Koide constant $2/3$ decomposes as $(1 + a^2/2)/N$ evaluated at $N = 3$ (generation count) and $a^2 = 2$ (Koide amplitude); the general formula is a trigonometric identity valid for $N \geq 3$

Method: Trigonometric identity for N equally-spaced objects on a circle; controlled-precision rational arithmetic for m_τ prediction; explicit separation of generation count contribution (denominator) from amplitude contribution (numerator)

Key Findings: General formula $(1+a^2/2)/N$ for arbitrary $N \geq 3$ and amplitude a ; the constant decomposes into two contributions neither of which alone produces $2/3$; $a = \sqrt{2}$ is the critical amplitude equivalent to the Koide relation; m_τ predicted at 1776.97 MeV (0.91σ from PDG); parameter count $18 \rightarrow 17$ conditional on formula being exact

Assumptions: $N = 3$ (empirical), equal spacing (maximum symmetry), $\sqrt{\text{inertia}}$ space (empirical), $a = \sqrt{2}$ (equivalent to the formula, not independently derived)

Does Not Claim: Derivation of the Koide formula itself; independent derivation of $a = \sqrt{2}$; validity of the identity for $N = 2$; applicability to quarks; VP $2/3$ connection is proven

Open Questions: Why equal spacing in $\sqrt{\text{inertia}}$ space; why $\sqrt{\text{inertia}}$ and not another power; what determines M and θ_0 ; whether the VP constant $2/3$ shares a common origin with the Koide constant $2/3$

Falsification: Five specific criteria including m_τ deviation $>3\sigma$, fourth-generation test of both $2/N$ and amplitude persistence, and derivation of equal spacing from deeper principle

APPENDIX G: THE GENERAL FORMULA AT ALL N AND SELECTED AMPLITUDES

Complete evaluation of $(1 + a^2/2)/N$ for $N = 2$ through 12 and five representative amplitudes.

N	$a = 0$ (degenerate)	$a = 1$ (moderate)	$a = \sqrt{2}$ (Koide/critical points)	$a = \sqrt{3}$ (one mass zero at 3 points)	$a = 2$ (one mass dominates)	Identity Valid?
2	1/2	3/4	1	5/4	3/2	NO — $\cos^2$ identity fails
3	1/3	1/2	2/3 ← Koide	5/6	1	Yes
4	1/4	3/8	1/2	5/8	3/4	Yes
5	1/5	3/10	2/5	1/2	3/5	Yes
6	1/6	1/4	1/3	5/12	1/2	Yes
7	1/7	3/14	2/7	5/14	3/7	Yes
8	1/8	3/16	1/4	5/16	3/8	Yes
9	1/9	1/6	2/9	5/18	1/3	Yes
10	1/10	3/20	1/5	1/4	3/10	Yes
11	1/11	3/22	2/11	5/22	3/11	Yes
12	1/12	1/8	1/6	5/24	1/4	Yes

Pattern at $a^2 = 2$ (Koide amplitude): The ratio is always $2/N$. This is the simplest column — the amplitude that produces the most compact formula. At every other amplitude, the ratio has a more complex numerator ($3/2N$ or $5/2N$). The Koide amplitude is the one where the general formula simplifies maximally.

Pattern at $a = 0$ (degenerate masses): The ratio is $1/N$, which is the Cauchy-Schwarz lower bound. All masses equal. No mass splitting.

Pattern at $a = 2$: The ratio is 1, which is the theoretical upper bound as one mass dominates. At $a = 2$, one of the three factors ($1 + 2\cos(\varphi)$) reaches -1 , producing $m = M^2(-1)^2 = M^2$ while another reaches 3, producing $m = 9M^2$. The mass hierarchy is 9:1:1.

APPENDIX H: THE CRITICAL AMPLITUDE — BOUNDARY ANALYSIS

At $a = \sqrt{2}$, the factor ($1 + \sqrt{2}\cos(\varphi)$) can equal zero. This table shows the mass distribution at $a = \sqrt{2}$ for the Koide parametrization as a function of θ_0 .

θ_0 (degrees)	$1 + \sqrt{2}\cos(\theta_0)$	$1 + \sqrt{2}\cos(\theta_0 + 120^\circ)$	$1 + \sqrt{2}\cos(\theta_0 + 240^\circ)$	Masses physical?	Zero mass?
0	2.414	0.293	0.293	Yes	No
30	2.225	-0.052	0.828	No — negative factor	—
45	1.999	-0.225	1.225	No	—
90	0.999	0.293	1.707	Yes	No
120	0.293	1.000	1.707	Yes	No (but close to zero)
132.7 (Koide)	0.052	1.264	1.684	Yes	Nearly zero (electron)
135	0.000	1.293	1.707	Boundary	Yes — one mass exactly zero
150	-0.225	1.518	1.707	No	—
180	-0.414	1.707	1.707	No	—

The Koide angle $\theta_0 \approx 132.7^\circ$ sits 2.3° below the critical angle 135° where one mass vanishes. The electron mass is small but nonzero because θ_0 is slightly below the critical angle. At $\theta_0 = 135^\circ$, the electron would be massless. At $\theta_0 > 135^\circ$, the parametrization produces a negative factor for the lightest species, which is unphysical.

The physical range of θ_0 at $a = \sqrt{2}$ is approximately $[-15^\circ, 135^\circ]$ modulo 120° (the three

equivalent orientations under permutation). The Koide angle sits near the upper boundary of this range, producing the extreme mass hierarchy $m_e \ll m_\mu \ll m_\tau$.

APPENDIX I: THE KOIDE PARAMETRIZATION — EXPLICIT VALUES

The three charged lepton masses extracted from the Koide parametrization $\sqrt{m_i} = M(1 + \sqrt{2} \cdot \cos(\theta_0 + 2\pi i/3))$ with M and θ_0 fitted to m_e and m_μ .

Parameter	Value	Source
M (scale)	$313.86 \text{ MeV}^{1/2}$	Fitted from m_e , m_μ
M^2	$98,508 \text{ MeV} = 98.5 \text{ MeV}$ (in mass units)	—
θ_0	$132.70^\circ = 2.316 \text{ rad}$	Fitted from m_e , m_μ
Lepton	i	$\varphi_i = \theta_0 + 2\pi i/3$
—	—	—
e	0	132.70°
μ	1	252.70°
τ	2	12.70°

Verification:

Check	Computed	Expected	Match?
$\Sigma \sqrt{m} / (3M)$	$(0.0410 + 0.5810 + 2.3797) / 3 = 1.0006$	1.0000 (cosine sum = 0 identity)	Yes — within truncation
$\Sigma m / (\Sigma \sqrt{m})^2$	$(0.511 + 105.66 + 1776.97) / (0.715 + 10.279 + 42.154)^2 = 0.66663$	$1882.14 / (53.148)^2 = 1882.14 / 2824.7 = 0.66663$	$2/3 = 0.66667$ — within truncation
m_e (predicted vs measured)	0.5110	0.51100	Input
m_μ (predicted vs measured)	105.66	105.658	Input
m_τ (predicted vs PDG)	1776.97	1776.86 ± 0.12	0.91σ

APPENDIX J: THE QUADRATIC SOLUTION — STEP BY STEP

The complete derivation of m_τ from the Koide formula treated as exact.

Step	Expression	Value
1. Define	$A = \sqrt{m_e} + \sqrt{m_\mu}$	$0.71485 + 10.2788 = 10.9936 \text{ MeV}^{(1/2)}$
2. Define	$B = m_e + m_\mu$	$0.51100 + 105.658 = 106.169 \text{ MeV}$
3. Let	$s = \sqrt{m_\tau}$ (unknown)	To be solved
4. Koide equation	$(B + s^2) / (A + s)^2 = 2/3$	—
5. Cross-multiply	$3(B + s^2) = 2(A + s)^2$	—
6. Expand RHS	$3B + 3s^2 = 2A^2 + 4As + 2s^2$	—
7. Rearrange	$s^2 - 4As + (3B - 2A^2) = 0$	Standard quadratic in s
8. Coefficients	a coeff = 1, b coeff = $-4A = -43.975$, c coeff = $3B - 2A^2 = 318.51 - 241.72 = 76.79$	—
9. Discriminant	$\Delta = 16A^2 - 4(3B - 2A^2) = 16A^2 - 12B + 8A^2 = 24A^2 - 12B$	—
10. Evaluate	$\Delta = 24 \times 120.86 - 12 \times 106.17 = 2900.6 - 1274.0 = 1626.6$	—
11. $\sqrt{\Delta}$	40.331	—
12. Two roots	$s = (43.975 \pm 40.331) / 2$	—
13. Positive root	$s_+ = 84.306 / 2 = 42.153 \rightarrow m_\tau = s_+^2 = 1776.9 \text{ MeV}$	Koide prediction
14. Negative root	$s_- = 3.644 / 2 = 1.822 \rightarrow m_- = s_-^2 = 3.32 \text{ MeV}$	No known particle

In Fraction arithmetic: Every step above is performed with Python Fractions. The square roots ($\sqrt{m_e}$, $\sqrt{m_\mu}$, $\sqrt{\Delta}$) are computed via Newton's method on Fractions with controlled simplification. The final m_τ Fraction has a truncation error bounded at 10^{-37} in the Koide ratio.

APPENDIX K: THE MASS HIERARCHY FROM THE PARAMETRIZATION

The Koide parametrization produces the extreme mass hierarchy $m_e : m_\mu : m_\tau \approx 1 : 207 : 3477$ from two parameters (M, θ_0) and the geometric structure (equal spacing, critical amplitude).

Ratio	From Parametrization	From Measurement	Match?
m_μ / m_e	$(1 + \sqrt{2} \cdot \cos(252.7^\circ))^2 / (1 + \sqrt{2} \cdot \cos(132.7^\circ))^2 = (0.581/0.041)^2 = 200.8$	206.768	3% — limited by θ_0 precision
m_τ / m_μ	$(1 + \sqrt{2} \cdot \cos(12.7^\circ))^2 / (1 + \sqrt{2} \cdot \cos(252.7^\circ))^2 = (2.380/0.581)^2 = 16.78$	16.817	0.2%
m_τ / m_e	Product of above = 3371	3477	3%
$\sqrt{(m_\tau / m_e)}$	$2.380/0.041 = 58.0$	58.97	1.6%
$\sqrt{(m_\mu / m_e)}$	$0.581/0.041 = 14.17$	14.38	1.5%

The hierarchy is geometric, not arithmetic. The three masses are not evenly spaced in mass. They are evenly spaced in angle on a circle in $\sqrt{m}$ space. The equal angular spacing (120°) combined with the critical amplitude ($a = \sqrt{2}$) and the specific orientation ($\theta_0 \approx 132.7^\circ$) produces the observed hierarchy. Moving θ_0 toward 135° would make the electron massless and the hierarchy infinite. Moving θ_0 toward 120° would equalize two masses (μ and e merging). The observed θ_0 sits near the critical boundary, producing the extreme but finite hierarchy.

APPENDIX L: THE $N = 2$ FAILURE — DETAILED

Why the general formula does not work for two objects.

Property	$N \geq 3$	$N = 2$
$\sum \cos(\theta + 2\pi k/N) = 0$	Yes — geometric series identity	Yes — $\cos(\theta) + \cos(\theta + \pi) = 0$
$\sum \cos^2(\theta + 2\pi k/N) = N/2$	Yes — double-angle cosines also cancel	No — $\cos^2(\theta) + \cos^2(\theta + \pi) = 2\cos^2(\theta)$, depends on θ
M and θ_0 cancel from ratio?	Yes — both sums depend only on N and a	No — ratio depends on θ_0
Formula $(1+a^2/2)/N$ applies?	Yes	No

Physical consequence: If two objects (e.g., up and down quarks, or electron and muon alone) are parametrized on a circle, the Koide-type ratio depends on their angular separation, not just on N and a . The formula does not predict a universal constant for doublets. This is consistent with the observation that no Koide-type relation holds for quark doublets or lepton doublets.

Mathematical detail for $N = 2$:

$$\begin{aligned} & \Sigma \cos^2(\theta + \pi k) \text{ for } k = 0, 1: \\ &= \cos^2(\theta) + \cos^2(\theta + \pi) \\ &= \cos^2(\theta) + \cos^2(\theta) \text{ [since } \cos(\theta + \pi) = -\cos(\theta), \text{ and } (-\cos(\theta))^2 = \cos^2(\theta)] \\ &= 2\cos^2(\theta) \end{aligned}$$

This equals $N/2 = 1$ only when $\cos^2(\theta) = 1/2$, i.e., $\theta = \pi/4$ or $3\pi/4$. For generic θ , the identity fails. The formula produces a θ -dependent ratio, not a universal constant.

APPENDIX M: QUARK SECTOR — WHY KOIDE FAILS

The Koide formula does not hold for quarks. This table documents the test.

Sector	Masses (MeV)	$\sqrt{m}$ values	Σm	$(\Sigma \sqrt{m})^2$	Ratio	2/3?
Charged leptons (e, μ , τ)	0.511, 105.66, 1776.86	0.715, 10.279, 42.153	1883.0	2824.7	0.6666	Yes (0.0009%)
Up-type quarks (u, c, t)	2.16, 1270, 172690	1.470, 35.64, 415.6	173962	204870	0.849	No
Down-type quarks (d, s, b)	4.67, 93.4, 4180	2.161, 9.664, 64.65	4278	5861	0.730	No
All quarks (u, d, s, c, b, t)	All six	All six	178240	—	—	Not applicable (N=6 formula would give $2/6 = 1/3$)
Neutrinos (ν_1 , ν_2 , ν_3)	~0.001, ~0.009, ~0.05 (eV)	Very uncertain	—	—	—	Insufficient precision to test

Why quarks fail:

Possible Reason	Explanation	Testable?
Equal spacing doesn't hold for quarks	Quarks are not at 120° in $\sqrt{m}$ space	Yes — compute angles and check
Different amplitude for quarks	$a \neq \sqrt{2}$ for quark sectors	Yes — fit a from quark masses
QCD corrections	Strong interaction renormalizes quark masses; Koide might hold at a specific scale	Partially — requires running masses to a common scale
Different symmetry structure	Quarks have color; leptons don't	Theoretical
The formula is specific to leptons	No deeper principle extends it	Possible — would limit the generalization

Fitted quark amplitudes:

Sector	Fitted a^2	$(1+a^2/2)/3$	Matches measured ratio?
Up-type (u, c, t)	3.55	0.924	No — measured ratio is 0.849, off by 9%
Down-type (d, s, b)	2.38	0.730	Fits by construction

The fit fails for up-type quarks because the three up-type masses are not well-described by equal spacing on any circle. The top quark mass dominates so heavily (172,690 vs 1,270 vs 2.16) that the parametrization cannot accommodate it at any a with equal spacing. The down-type quarks can be fitted but the amplitude ($a^2 = 2.38$) differs from the Koide value ($a^2 = 2$), and the fit quality is poor.

APPENDIX N: THE TWO CONTEXTS OF $2/3$ — DETAILED COMPARISON

Property	Koide $2/3$	Vacuum Polarization $2/3$
Formula	$(\Sigma m)/(\Sigma \sqrt{m})^2 = 2/3$	Subtracted VP constant = $-2/3$
Decomposition	$(1 + a^2/2)/N$ at $N = 3$, $a^2 = 2$	$-5/3 + 1 = -2/3$ (Feynman integral minus subtraction)

Property	Koide 2/3	Vacuum Polarization 2/3
The 3	Three charged lepton generations	3^2 in denominator of integral $\int x(1-x)\ln(x)dx = -5/36$; $36 = 4 \times 9 = 4 \times 3^2$
The 2	$1 + a^2/2 = 1 + 1 = 2$ from critical amplitude	$(5/3 - 1) = 2/3$; the 2 appears as the subtraction-corrected numerator
The 5 (absent in Koide)	Does not appear	$5 = 3^2 - 2^2$ from Feynman parameter integral
Physical context	Mass relation among three generations	Quantum correction to photon propagator from one fermion loop
Number of fermion species involved	3 (e, μ , τ)	1 per loop (any charged fermion)
Why 2/3	Generation count $\times$ critical amplitude	Integral of momentum-sharing probability $\times$ spin $\times$ subtraction
Same origin?	Open question	Open question

What would connect them: A derivation showing that the Feynman parameter integral for a loop of three equally-spaced fermions (in some appropriate space) produces the same 2/3 as the Koide formula would connect the two contexts. No such derivation exists. The coincidence may be a consequence of 2/3 being a common small rational that appears in many mathematical contexts.

What would distinguish them: If the Koide formula holds for a different number of generations (e.g., $N = 4$ with ratio $2/4 = 1/2$) while the VP constant remains $-2/3$ regardless of generation count (which it does — the VP constant is per-fermion and doesn't depend on the number of species), the two 2/3's would be unrelated. The VP constant is independent of N . The Koide constant depends on N . This difference argues against a common origin.

APPENDIX O: THE CAUCHY-SCHWARZ BOUNDS — COMPLETE

The Koide ratio $(\Sigma m)/(\Sigma \sqrt{m})^2$ is bounded by the Cauchy-Schwarz inequality and its inverse.

Bound	Value (N=3)	Condition	Physical Meaning	Koide Position
Lower bound (Cauchy-Schwarz)	$1/N = 1/3$	All masses equal: $m_e = m_\mu = m_\tau$	Zero mass splitting; degenerate generations	Below Koide by $1/3$
Koide value	$2/3$	$a = \sqrt{2}$, equal spacing	Critical amplitude; masses can reach zero	Exact midpoint
Upper bound (one mass dominates)	$\rightarrow 1$	One mass $\rightarrow \infty$, others fixed	Complete hierarchy; one generation dominates	Above Koide by $1/3$
Distance from lower to Koide	$2/3 - 1/3 = 1/3$	—	—	—
Distance from Koide to upper	$1 - 2/3 = 1/3$	—	—	—
Midpoint of $[1/3, 1]$	$(1/3 + 1)/2 = 2/3$	—	—	Koide IS the midpoint

The midpoint property: The Koide value $2/3$ sits exactly at the midpoint of the allowed range $[1/3, 1]$. This is not an independent observation — it follows directly from $a^2 = 2$. At $a = 0$ (lower bound), the ratio is $1/N$. At $a \rightarrow \infty$ (upper bound), the ratio $\rightarrow 1$. The midpoint occurs when $(1 + a^2/2)/N = (1/N + 1)/2$, which gives $a^2 = 2(N-1)/N$. For $N = 3$: $a^2 = 4/3$, not 2. So the midpoint of the allowed range and $a^2 = 2$ are NOT the same condition — the exact midpoint would require $a^2 = 4/3$, not $a^2 = 2$.

Correction to the midpoint claim: The Koide value $2/3$ is the midpoint of $[1/3, 1]$ arithmetically. But the Cauchy-Schwarz upper bound for three positive masses approaching the limit is strictly less than 1 (it equals 1 only in the degenerate limit of one mass infinite). For finite masses, the actual achievable range depends on the specific mass ratios. The midpoint observation is noted as approximate, not exact.

APPENDIX P: SENSITIVITY OF THE KOIDE PREDICTION TO INPUT MASSES

How the predicted m_τ changes with small variations in the input masses m_e and m_μ

Variation	m_e (MeV)	m_μ (MeV)	Predicted m_τ (MeV)	Shift from baseline	Koide ratio
Baseline	0.51099895	105.6583755	1776.97	0	2/3 (exact)
$m_e + 1\sigma$	0.51099910	105.6583755	1776.97	+0.001	2/3
$m_e - 1\sigma$	0.51099880	105.6583755	1776.97	-0.001	2/3
$m_\mu + 1\sigma$	0.51099895	105.6583778	1776.98	+0.01	2/3
$m_\mu - 1\sigma$	0.51099895	105.6583732	1776.96	-0.01	2/3
$m_e + 10\sigma$	0.51100045	105.6583755	1776.98	+0.01	2/3
$m_\mu + 10\sigma$	0.51099895	105.6583985	1777.06	+0.09	2/3

The prediction is stable. m_τ is insensitive to m_e variations (the electron mass is so small that even 10σ shifts change m_τ by only 0.01 MeV). m_τ is moderately sensitive to m_μ (10σ shift in m_μ changes m_τ by 0.09 MeV). In all cases, the Koide ratio remains exactly 2/3 by construction — the prediction is the solution of the Koide equation, so the ratio is 2/3 for any input masses.

The dominant experimental uncertainty in the m_τ prediction comes from m_μ , not m_e . But m_μ is known to 10 significant figures, so the prediction uncertainty from input masses is below 0.01 MeV — far smaller than the PDG uncertainty on m_τ itself (0.12 MeV). The test of the formula is limited by the m_τ measurement, not by the m_e or m_μ inputs.

APPENDIX Q: THE PARAMETER REDUCTION — COMPLETE ACCOUNTING

Step	Parameter Count	What Changed	Paper	Method	Conditional?
Standard Model (textbook)	19	Baseline: 3 gauge couplings, 6 Yukawa couplings, Higgs μ^2 and λ , 3 CKM angles, 1 CKM phase, θ QCD, plus 3 neutrino masses and 3 PMNS angles if massive neutrinos	—	—	—
After PHYS-7	18	θ QCD = 0 derived as ground state	PHYS-7	Energy minimization of integer-topological system	No — the ground state argument is unconditional
After PHYS-8	17	m_τ derived from m_e , m_μ via Koide	PHYS-8	Trigonometric identity at $N = 3$, $a^2 = 2$	Yes — conditional on Koide formula being exact
Future: if Koide extends to quarks	15 or less	m_b from m_d , m_s ; m_t from m_u , m_c	Hypothetical	Same formula at different amplitude?	Highly conditional — quarks don't currently fit
Future: if M or θ_0 derived	16	One of $\{m_e, m_\mu\}$ derived from the other + structure	Hypothetical	Would require deriving M or θ_0 from deeper principle	Speculative

Step	Parameter Count	What Changed	Paper	Method	Conditional?
Theoretical minimum	?	All parameters derived from structure	ToE	Unknown	Unknown

Current status: 19 → 18 (unconditional) → 17 (conditional on Koide exactness at 0.91σ).

APPENDIX R: FUTURE m_τ MEASUREMENTS — WHAT WOULD CONFIRM OR FALSIFY

Experiment/Facility	Expected m_τ Precision	Timeline	What It Would Show
Belle II (KEK)	$\pm 0.05\text{-}0.08$ MeV	2025-2030	If central value moves toward 1776.97: strengthens Koide to $>2\sigma$ confirmation. If moves below 1776.74: $>2\sigma$ tension with Koide.
FCC-ee (CERN, proposed)	$\pm 0.01\text{-}0.02$ MeV	2040s	Definitive test. At this precision, Koide prediction of 1776.97 is either confirmed (within 0.02) or falsified ($>5\sigma$ tension if PDG value holds).
Super Tau-Charm Factory (proposed)	$\pm 0.03\text{-}0.05$ MeV	2030s	Intermediate precision. Could confirm or begin to exclude.
Current PDG average	± 0.12 MeV	Now	0.91σ — consistent with exact Koide

The falsification threshold: At the current 0.12 MeV uncertainty, the Koide prediction would be excluded at 3σ if $m_\tau < 1776.61$ MeV or $m_\tau > 1777.33$ MeV. The current central value 1776.86 is within 1σ . Improvement to ± 0.05 MeV would narrow the 3σ window to 1776.82-1777.12, still including the Koide prediction. Improvement to ± 0.02 MeV would make the test sharp: the 3σ window would be 1776.91-1777.03, centered on the Koide prediction.

[[HOWL-PHYS-8-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-8-2026>

[[HOWL-PHYS-1-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026>

[[HOWL-PHYS-5-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-5-2026>

[[HOWL-PHYS-7-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-7-2026>

[[HOWL-MATH-1-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-1-2026>

[[HOWL-MATH-2-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-2-2026>

[[HOWL-MATH-3-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-3-2026>

[[HOWL-PHYS-2-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-2-2026>

[[HOWL-PHYS-3-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-3-2026>

[[HOWL-PHYS-4-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-4-2026>

[[HOWL-PHYS-6-2026](#)] Paternal Operationalism. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-6-2026>